Integrating Lean Six Sigma Indicators into Business Intelligence Systems: A Systematic Review across Sectors and Metrics

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***Abstract*— This study systematically reviews the use of Lean Six Sigma (LSS) performance indicators within Business Intelligence (BI) environments between 2020 and 2025, with specific emphasis on manufacturing applications. From 104 Scopus-indexed Q1–Q2 journal articles, only seven reported quantifiable LSS indicators—such as Process Cycle Efficiency (PCE), Cost of Poor Quality (COPQ), Return on Investment (ROI), defect rate, and customer satisfaction—and discussed their analytical relevance. These indicators were extracted and mapped to BI analytical functions and DMAIC phases to identify utilization patterns. The review finds that, although LSS metrics are widely reported, system-level integration through BI or ERP architecture remains underdocumented, particularly concerning data pipelines, computation layers, and dashboard mechanisms. As its key contribution, this study proposes a dual taxonomy that classifies LSS indicators by BI function and DMAIC stage. Findings show that manufacturing contexts exhibit the highest ERP–BI maturity, forming the empirical basis for an integrated model that supports real-time, enterprise-wide performance analytics.**

Keywords—Lean Six Sigma, Business Intelligence, Key Performance Indicators, Systematic Literature Review, ROI, COPQ

I. Introduction

Lean Six Sigma (LSS) is a structured methodology widely applied to improve process efficiency and quality by eliminating waste and reducing operational variation [1][2]. Integrating Lean's waste minimization with Six Sigma's variation control, LSS has become a core foundation for continuous improvement initiatives across multiple sectors. Although manufacturing remains its primary domain, LSS has also been adopted in services, healthcare, government, and education, demonstrating its adaptability across diverse organizational contexts [3][4].

In recent years, research has increasingly emphasized the quantitative outcomes of LSS implementations. Operational indicators such as Process Cycle Efficiency (PCE), Cost of Poor Quality (COPQ), and Return on Investment (ROI) have been widely used to assess project effectiveness and justify improvement investments [5][6][7]. These metrics not only provide evidence of technical improvement but also play a crucial role in enabling organizations to make data-driven strategic decisions and sustain competitive advantage [8][9].

A preliminary screening of 40 high-quality (Q1–Q2) articles published between 2020 and 2025 indicates growing interest in embedding LSS outcomes into broader performance-monitoring mechanisms, particularly within Business Intelligence (BI) environments. Studies by Adeodu (2021), Persis et al. (2022), and Widiwati et al. (2024) illustrate that indicators such as defect reduction, PCE improvement, and customer satisfaction are increasingly used as formal components in enterprise performance evaluation—especially in discrete manufacturing settings where digital performance systems are more mature [7][10][11].

In parallel, organizations are increasingly investing in BI systems to facilitate data-driven management. While some studies have documented the integration of LSS-driven improvements into performance dashboards, the literature remains fragmented. It lacks a consistent mapping of which LSS metrics are utilized, how they are incorporated into BI systems, and in what contexts they deliver value. Business Intelligence (BI) refers to a technological and analytical ecosystem comprising data integration, data warehousing, dashboards, and analytical tools that transform operational data—primarily sourced from ERP or production systems into actionable insights for decision-making [12][13]. In modern manufacturing, BI serves as an analytical layer above transactional systems, providing real-time visibility into operational performance, financial outcomes, and customer-focused metrics. Integrating LSS indicators into BI platforms, therefore, strengthens the organization's analytical capability and supports responsive decision-making in dynamic production environments [14][15].

Despite this emerging development, the literature remains fragmented. Only seven of the articles reviewed explicitly describe the use of LSS indicators within BI dashboards or analytics systems, and even fewer provide detailed explanations of how these indicators are computed, which BI functions they support, or how they align with continuous improvement methodologies. Studies such as Gupta et al. (2020) and Hundal et al. (2020) discuss LSS in relation to Big Data and digitalization; however, they do not elaborate on functional integration within BI architectures [16][17]. This gap suggests the absence of a structured framework that links LSS performance metrics with enterprise-level analytics—particularly in manufacturing contexts where ERP–BI ecosystems are well established.

This disconnect highlights a critical opportunity to bridge continuous improvement and real-time analytics by systematically identifying and categorizing the LSS indicators used within BI platforms. However, existing studies provide limited guidance on how specific LSS metrics correspond to BI analytical functions such as operational monitoring, financial evaluation, or customer insight visualization.

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Therefore, this study aims to systematically classify LSS performance indicators explicitly utilized in Business Intelligence contexts between 2020 and 2025. It introduces a dual taxonomy that (1) categorizes LSS indicators according to BI analytical functions and (2) maps their presence across DMAIC phases and industrial sectors—particularly discrete manufacturing. This contribution provides a structured reference for researchers, practitioners, and system architects seeking to align LSS metrics with enterprise-level analytics and digital performance management systems..

II. METHODOLOGY

This study employs a semi-systematic, content-mapping approach designed to capture not only publication patterns but also the functional use of Lean Six Sigma (LSS) performance indicators within Business Intelligence (BI) contexts. A semi-systematic strategy is appropriate for this topic because prior studies rarely describe BI integration in formal terms, requiring manual extraction and mapping of indicators across BI analytical functions and DMAIC phases.

The literature search was conducted in the Scopus database using the query TITLE-ABS-KEY (“Lean Six Sigma”) AND TITLE-ABS-KEY (“case study”), restricted to English-language publications from 2020 to 2025 and journal articles. This search produced 104 records. Following the PRISMA 2020 protocol, the review progressed through four stages: identification, screening, eligibility, and inclusion. After title–abstract screening, 64 records remained for further assessment, and 40 full texts were retrieved. Of these, 20 articles satisfied the basic inclusion criteria (journal, English, complete publication). Subsequent content evaluation identified 13 articles that reported quantifiable LSS outcomes—such as Process Cycle Efficiency (PCE), Cost of Poor Quality (COPQ), Return on Investment (ROI), defect rate, and customer satisfaction—based on real operational data, predominantly in manufacturing settings. Finally, seven studies explicitly utilized LSS indicators within BI dashboards, reporting systems, or analytics environments and were therefore included in the qualitative synthesis.

This multi-stage filtering process, summarized in the PRISMA diagram (Figure 1), ensures transparency and reproducibility. All retained articles were appraised using a simplified AMSTAR-2 quality protocol, evaluating methodological rigor, clarity of indicator reporting, and relevance to ERP–BI–LSS integration. Inter-reviewer agreement reached a kappa value of 0.87, indicating strong reliability. This rigorous screening ensures that the final subset of seven studies offers a robust empirical foundation for the dual taxonomy developed in this review, enabling the structured mapping of LSS performance indicators across BI analytical functions and DMAIC phases. The resulting synthesis supports the formulation of the integrated data-driven performance framework presented in the Results section.

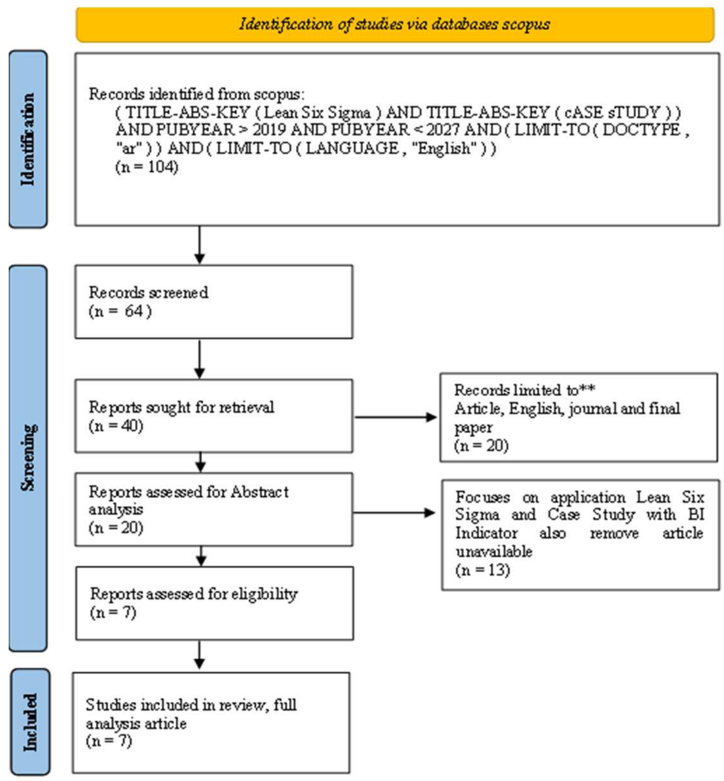

Fig. 1. PRISMA Diagram in this Study

III. RESULT AND DISCUSSION

A bibliometric overview was conducted to examine publication dynamics and thematic concentrations between 2020 and 2025. As shown in Figure 2, research on Lean Six Sigma (LSS) and data-driven performance analytics shows a steady upward trend, with the most substantial increase occurring between 2022 and 2024. This pattern indicates growing academic attention toward integrating continuous improvement methodologies with enterprise-level analytics and digital monitoring systems. The most frequent thematic keywords across the reviewed articles include process efficiency, cost of poor quality, key performance indicators, operational performance, and data-driven management. These recurring terms suggest a clear shift in the literature—from conventional, project-based LSS applications toward digitalized performance measurement supported by real-time analytics. This thematic evolution reinforces the relevance of the dual taxonomy developed in this study, which systematically maps LSS indicators to BI analytical functions and DMAIC phases.

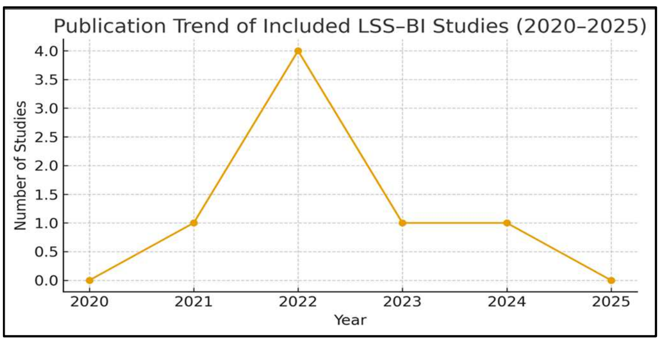

Fig. 2. Publication trend of including LSS-BI Studies

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Figure 2 presents the publication trend of the seven included LSS–BI studies between 2020 and 2025. The most substantial growth occurred in 2022, followed by a gradual decline, reflecting the sporadic yet emerging nature of LSS–BI integration research. This pattern aligns with broader interest in digitalized performance measurement, as noted in the bibliometric overview, where keywords such as process efficiency, cost of poor quality, key performance indicators, and data-driven management increasingly appeared in recent publications. These trends reinforce the shift from traditional, project-centric LSS applications towards enterprise-level analytics supported by real-time monitoring.

Sectoral analysis shows that 57% of the reviewed studies originate from manufacturing, followed by logistics (29%) and healthcare (14%), as illustrated in Figure 3. The dominance of manufacturing is consistent with its higher ERP maturity, richer operational datasets, and established performance measurement culture, all of which enable the extraction and visualization of LSS indicators within BI environments. In contrast, logistics and healthcare studies focus more on service quality, patient satisfaction, and cost reduction, which align with customer-oriented KPIs rather than system-level operational analytics.

Across the seven included studies, 12 unique LSS indicators were identified. The most frequently reported metrics were Process Cycle Efficiency (PCE), Cost of Poor Quality (COPQ), Return on Investment (ROI), Defect Rate, Lead Time, Waste Reduction, and Customer Satisfaction. Following the analytical classification in Table 2, these indicators map onto four BI functional categories:

- Process Monitoring & Control: PCE, Defect Rate, Quality Rate
- Financial Evaluation: ROI, COPQ, Cost Reduction
- Operational Visualization: Lead Time, Waste Reduction
- Customer Response & Insights: Customer Satisfaction

This mapping demonstrates that LSS indicators are distributed across multiple BI modules rather than concentrated within a single analytical function. It also reflects how the role of LSS metrics extends beyond project-level validation into broader enterprise reporting. Furthermore, aligning these indicators with DMAIC phases (Table 3) shows that:

- PCE and Defect Rate are frequently used to support the Measure and Control phases.
- COPQ and Quality Rate are commonly used in the Analyze phase.
- Lead Time and Waste Reduction primarily appear in the Improve phase.
- ROI and Customer Satisfaction are typically associated with Control, representing sustained performance verification.

This dual mapping—across BI analytical functions and DMAIC phases provides a more granular understanding of how LSS indicators contribute to enterprise analytics. The synthesis of these patterns forms the conceptual foundation for the integrated ERP–LSS–BI framework presented in Figure 6, which positions LSS indicators as analytically derived metrics within BI environments to support data-driven operational, tactical, and strategic decision-making.

Despite increasing evidence of LSS–BI convergence across the reviewed studies, none of them present a fully specified architecture detailing how LSS indicators are generated from ERP or shop-floor data and subsequently processed within BI environments. Essential system-level components such as data extraction pipelines, indicator computation stages, visualization layers, and interoperability between ERP and BI modules—remain largely undocumented in the existing literature. To address this gap, the present study introduces a dual taxonomy framework (Figure 6) that links LSS indicators to both BI analytical functions and DMAIC phases. This structure clarifies where each indicator is operationalized within enterprise analytics and establishes a foundational model for future development of automated, real-time performance dashboards within ERP–BI ecosystems.

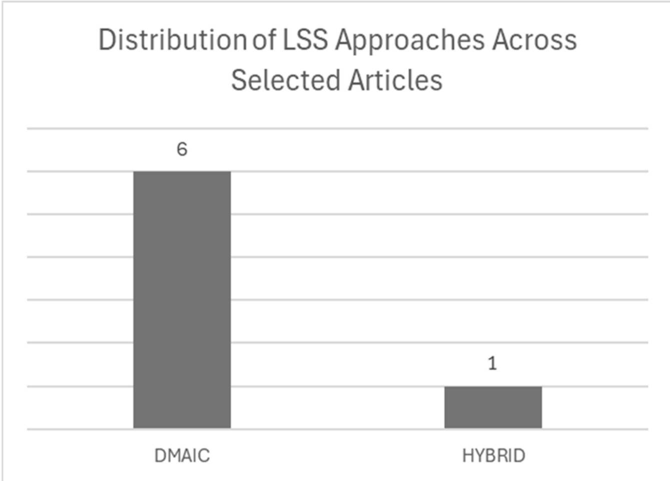

Fig. 3. Distribution of LSS Approaches Across Selected Articles

Figure 4 presents the industrial distribution of the seven included studies, showing that manufacturing accounts for 57%, followed by logistics (29%) and healthcare (14%). This distribution reinforces the earlier finding that manufacturing sectors exhibit greater ERP maturity and more comprehensive performance documentation, enabling stronger integration of LSS indicators within BI environments. Figure 5 further demonstrates the alignment between LSS indicators and BI analytical functions, as shown in Table 2. Financial indicators such as ROI and COPQ align with Financial Evaluation, while PCE, Defect Rate, and Quality Rate support Process Monitoring & Control. Lead Time and Waste Reduction map onto Operational Visualization, and Customer Satisfaction aligns with Customer Response & Insights. These cross-functional mappings highlight that LSS indicators are not limited to project-level monitoring but can be embedded across multiple BI modules to support enterprise-wide analytics.

Theoretically, the dual taxonomy proposed in this review advances the understanding of metric–function alignment by demonstrating how LSS indicators can be operationalized across BI analytical functions and DMAIC stages. In practice, taxonomy provides actionable guidance for system architects and quality managers seeking to embed LSS metrics into BI dashboards for real-time operational monitoring, financial evaluation, and customer satisfaction tracking. This review is limited to Scopus-indexed journal articles and excludes grey literature, potentially leading to an underrepresentation of emerging industrial implementations. Future research should expand database coverage to IEEE Xplore and Web of Science, incorporate co-citation and bibliographic coupling

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analyses, and develop empirical prototypes that embed LSS indicators into ERP–BI environments. Integrating predictive analytics or AI-driven modules would further enhance continuous monitoring and adaptive decision support, accelerating the transition toward data-driven operational excellence.

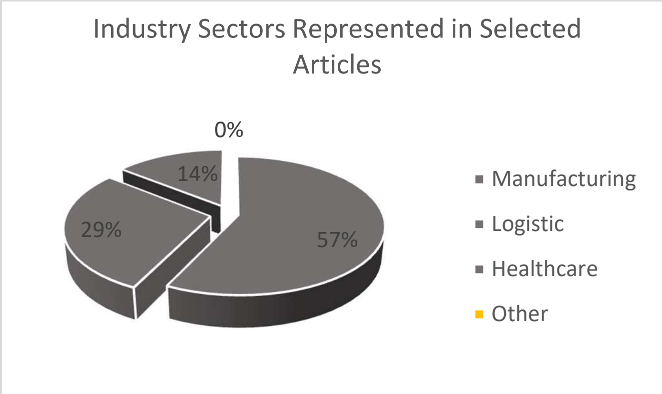

Fig. 4. Industry Sectors Represented in Selected Articles

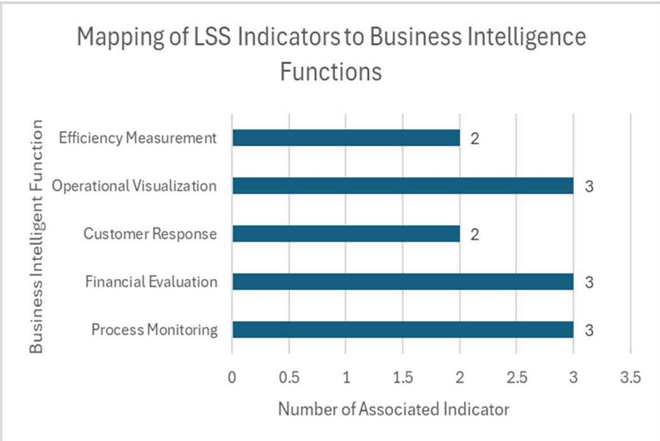

Fig. 5. Mapping LSS Indicator to Business Intelligence

TABLE I. SUMMARY OF REVIEWED ARTICLES RELEVANT TO BUSINESS INTELLIGENCE INTEGRATION.

Article Title	Key indicator	Methode	Key Outcomes
Adeodu (2021) [18]	ROI, COPQ, Reduce Manpower	DMAIC	Cost and labor efficiency
Adeodu et al. (2023) [5]	ROI, COPQ, Waste Reduction	DMAIC	Financial and process performance
Persis et al. (2022) [10]	Customer Satisfaction, PCE (%)	Hybrid	Customer responsiveness
Widiwati et al. (2024) [11]	ROI, PCE (%), Quality Rate	DMAIC	Process performance visualization
Panayiotou et al. (2022) [19]	Customer Satisfaction, Defect Rate	DMAIC	Quality and customer perception
Daniyan et al. (2022) [20]	Reduce Cost, COPQ, ROI	DMAIC	Cost efficiency
Nedra et al. (2022) [21]	PCE (%), Quality Rate, Reduce waste	DMAIC	Process-based monitoring

Across the reviewed studies, several authors reported improvements in LSS performance indicators, including reductions in COPQ, increases in ROI, improvements in Quality Rate, and higher Customer Satisfaction. For example, Adeodu [18] and Adeodu et al. [5] demonstrated positive financial outcomes from LSS initiatives, while Persis et al. [10] and Panayiotou et al. [19] gained in PCE, defect reduction, and customer-focused metrics. However, these studies consistently stopped at reporting performance outcomes and did not explain how such indicators were technically integrated into BI or ERP environments. Descriptions of dashboards, data flows, or visualization mechanisms were largely absent, and no study provided a system-level architecture that connects LSS indicators with BI modules.

Table I shows this fragmentation clearly: although indicators are mentioned, their analytical placement within BI systems—such as process monitoring, financial evaluation, operational visualization, or customer insights is rarely specified. Table II addresses this gap by classifying indicators according to BI analytical functions. Indicators such as PCE, Defect Rate, and Quality Rate align with process monitoring; ROI and COPQ with financial evaluation; Lead Time and Waste Reduction with operational visualization; and Customer Satisfaction with customer-insight functions. This classification suggests that LSS indicators are cross-functionally relevant and can serve as structured inputs to enterprise analytics when properly embedded in BI systems.

Table III further maps these indicators to DMAIC phases, revealing that most appear during the Improve and Control stages-where validated data is available for monitoring and optimization. Early-stage indicators, such as Lead Time or initial customer feedback, commonly emerge in the Define–Measure phases but are seldom carried forward into BI reporting. This indicates an underutilized opportunity: integrating DMAIC-aligned indicators into BI dashboards would enable organizations to monitor improvement trajectories rather than just reporting end-of-project results.

TABLE II. CLASSIFICATION OF LEAN SIX SIGMA INDICATORS BY BUSINESS INTELLIGENCE FUNCTION

Business Intelligence Functions	Relevant LSS Indicators
Process Performance Monitoring	PCE, Defect Rate, Quality Rate, Cycle Time
Financial Evaluation	ROI, COPQ, Cost Reduction
Operational Visualization	Lead Time, Waste Reduction
Customer & Market Response	Customer Satisfaction
Internal Efficiency	Reduce Manpower, Cycle Time

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TABLE III. LSS INDICATORS MAPPED TO DMAIC PHASES

DMAIC Step	Dominant Indicator
Define	Customer Satisfaction
Measure	COPQ, Lead Time, Defect Rate
Analyze	COPQ, Waste Reduction, Defect Rate
Improve	Lead Time, Waste Reduction, PCE
Control	PCE, Quality Rate, ROI

Despite this potential, literature uniformly lacks technical explanations of how LSS metrics flow from ERP or shop-floor systems into BI platforms. Essential elements such as ETL pipelines, indicator-calculation layers, visualization modules, and ERP–BI interoperability remain unreported. This gap validates the novelty of the present study. By synthesizing BI-function mappings (Table 2) and DMAIC-phase alignments (Table 3), this study develops the dual taxonomy needed to conceptualize how LSS indicators can be operationalized within enterprise analytics ecosystems.

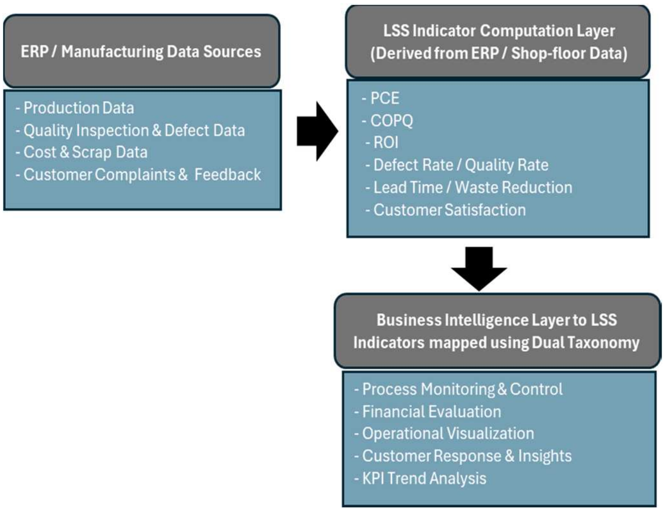

Fig. 6. Dual-Taxonomy-Based Framework for Operationalizing LSS Indicators in BI Environments

To complement these classifications, Figure 6 presents a conceptual framework that illustrates how LSS performance indicators, such as PCE, COPQ, ROI, Customer Satisfaction, etc., are computed from ERP or shop-floor data and subsequently processed by BI analytical modules. The ERP/manufacturing data layer represents the transactional records from which these indicators are derived, including production data, quality inspection logs, defect records, scrap and cost tracking, and customer complaints and feedback. Although none of the reviewed studies explicitly documented these data structures, positioning ERP as the foundational data source clarifies that LSS indicators are not standalone metrics but analytical outputs derived from enterprise systems. Building on this foundation, BI analytical functions operationalize the indicators to support multi-level decision-making, including operational monitoring, tactical performance evaluation, and strategic analysis. This framework directly addresses the missing architecture in the literature and provides a foundational model for developing integrated ERP–BI dashboards and future AI-enabled monitoring systems.

Although the proposed framework is conceptually generalizable, its empirical foundation is derived primarily from discrete manufacturing studies, which dominated the reviewed literature. Manufacturing environments typically exhibit higher ERP maturity, richer production and quality datasets, and more structured performance measurement routines. For this reason, the integrated ERP–LSS–BI architecture presented in Figure 6 is most directly applicable to manufacturing settings, where LSS indicators such as PCE, COPQ, defect rate, and scrap-related costs can be systematically extracted and embedded into BI analytical modules.

IV. CONCLUSION AND FUTURE WORK

Future research should focus on designing integrated system architectures that operationalize LSS indicators within real-time BI and ERP environments. This includes developing ETL pipelines, standardized indicator-calculation layers, and dynamic dashboards that visualize DMAIC-aligned metrics across operational, tactical, and strategic levels. Empirical prototyping, such as implementing LSS-based KPI models within manufacturing ERP systems, will be essential for validating the practicality and performance impact of such architectures.

In addition, broader bibliometric coverage (e.g., IEEE Xplore, Web of Science) and co-citation network analysis may uncover emerging patterns in BI–LSS integration. Incorporating predictive analytics or AI-enabled monitoring modules represents another promising direction, enabling organizations to track process deviations, forecast performance trends, and automate decision-support workflows. Such advancements would strengthen the convergence of continuous improvement and digital analytics, supporting the transition toward resilient, data-driven operational excellence.

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